PROJECT AND TEAM INFORMATION

Project Title

Smart Campus Lost and Found Matching System

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Students/staff tend to lose things like id cards, wallet, phone, keys, books bags, earphones etc in the campus. Mostly these lost items are found by someone but not sure who exactly lost them, hence it becomes hard to return them to the rightful owner. Currently the lost found items are shared on WhatsApp groups(class groups), notice boards or just asking around, making it a bit chaotic since losing track of what was given and who lost them and also there is no search feature The purpose of my Smart Campus Lost and Found system is to give an overview of what is being found and by whom and where the item was found and also a feature to search for a lost item and notify if found or not. It is crucial to make this a great project so it is easier to have your lost item returned to you. This project will also enable me to test my knowledge learnt in Data Structures using C and also Object Oriented Programming using C++.

State of the Art / Current solution

At Present the lost and found items in campus mainly relies upon WhatsApp groups, class groups, college notice boards or by enquiring to friends and staff. The found items on the campus are reported either by the students in any of the above-mentioned ways or directly to the college office or security. The students who lost the items had no choice but to check the groups or inquire on the campus regarding the found items. So a proper system was not available for the lost and found items in the college. Though this method works out fine, it may sometimes take a long time and become tedious. The main advantage of our project is that, it provides a single platform for the students to report and search for the lost and found items in the college.

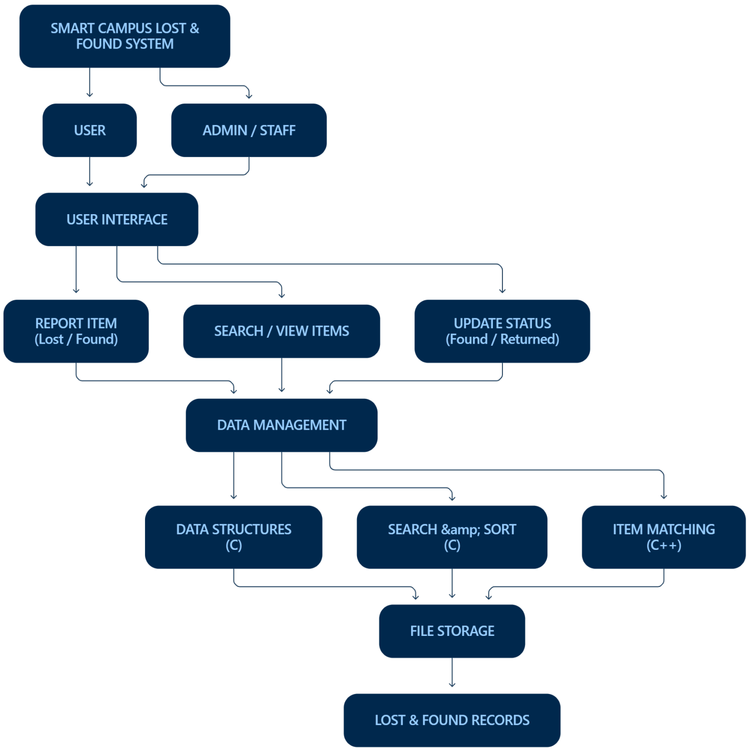
Project Goals and Milestones

The main objective of our project is to design a Smart Campus Lost and Found System that will help students and staff find lost items more efficiently. The system will enable users to report lost and found items using item name, category, location, date, and description. It will also offer searching options, item matching options, and status updates on located items. Our first milestones involve identifying the current processes and problems in the lost and found system. We will also establish the information required in the system and develop a model of the new system structure. We intend to use Data Structures using C to design and implement a storage management system. In addition, we will apply OOP concepts in C++ to develop the proposed system model. The next milestones consist of developing searching, sorting, and item-matching algorithms, testing and debugging the algorithms, and improving the system functions. We will also design and construct a working model of the project and conclude the documentation and report of the project. Overall, our project aims to improve the lost and found system on campus by making the process faster and more efficient.

Project Approach

We intend to develop the Smart Campus Lost and Found System as a simple application where students and staff report, search, match, and update lost and found items. Firstly, we will determine the features and the general flow of the system. We will develop Data Structures and concepts like searching, sorting, linked lists, stacks, and queues using C. We will apply C++ to develop OOP concepts such as classes, objects, inheritance, and encapsulation. Initially, the project will be developed as a console-based application using either the Visual Studio Code or Code::Blocks. We will also utilize file handling to store records and test the system under different scenarios to enhance its reliability and applicability.

System Architecture



Project Outcome / Deliverables

The main outcome of our project will be a working Smart Campus Lost and Found System that offers a simple and organized way to manage lost and found items on campus. Users will be able to add details of lost or found items, search for items, search for possible matches, and update the status when an item is returned. The project will also demonstrate the use of Data Structures in C such as linked lists, stacks, queues, searching, and sorting, as well as OOP concepts in C++ such as classes, objects, inheritance, and encapsulation. The main deliverables will include the completed working application, source code, file-based data storage, testing results, and project documentation. We will also prepare a final presentation and demonstration of the system. The final system should make it easier and faster for students and staff to report and recover lost belongings.

Assumptions

We assume that the students and staff will give truthful and helpful information when submitting lost or found We assume that the user will have access to a computer or device on which the application can be used. We assume that the program will be mostly used in the campus of the college, where the amount of records will be small enough to not strain the data structures. We assume that the user will update the status of the item once it has been found or returned. We assume that the person submitting the item will include enough information to possibly identify and match the item.

References

We referred to the following resources while planning and developing the project:
1. GeeksforGeeks – C Programming, Data Structures and C++: https://www.geeksforgeeks.org/
2. W3Schools – C and C++ basics: https://www.w3schools.com/
3. Programiz – C and C++ programming concepts: https://www.programiz.com/
4. C++ Reference – C++ language and OOP concepts: https://en.cppreference.com/
5. TutorialsPoint – Data Structures and C++ programming: https://www.tutorialspoint.com/
We also referred to our college lecture notes and class materials for Data Structures in C and Object-Oriented Programming in C++.